

YEHYA ELNOTI

AEROSPACE ENGINEERING
STUDENT

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SUMMARY

Aerospace engineering student with hands-on experience in autonomous UAV development. Experience in Python, C++, ground control station (GCS), and avionics integration. Background includes contributing to fixed-wing, VTOL and multirotor projects, with focus on autopilot control systems and mission planning. Experience in developing and testing software/hardware in simulation environments SITL/HITL, and preparation of simulation environment.

PUBLICATION

MATHEMATICAL MODELING OF QUADROTOR DRONE

Livre de Lyon · Mar 23, 2025 · Book: AI Applications in Electrical and Electronics Engineering

In this chapter, we explored the fundamental aspects of quadrotor UAV modeling and control architecture, emphasizing the importance of accurate mathematical representations in developing control strategies. By applying classical mechanics principles, particularly the Newton-Euler and Lagrange Euler methods, we developed a detailed nonlinear model that captures the complexities of quadrotor dynamics, including aerodynamic forces, actuator behavior, and environmental influences. This mathematical foundation is essential for designing control systems capable of managing the challenges posed by the quadrotor's underactuated nature, where six degrees of freedom must be controlled with only four inputs. Additionally, we explored the intricate relationship between the quadrotor's translational and rotational subsystems, highlighting the difficulties of maintaining stable flight and precise trajectory tracking.

SKILLS

- PX4-Autopilot/ArduPilot
- Python
- C++
- Software/Hardware in the loop (SITL/HITL)
- Flight Simulation (Gazebo/Xplane)
- Matlab Simulink
- Blender
- Solidworks (CSWA Certified)

EDUCATION

- ISTANBUL AYDIN UNIVERSITY
Aerospace Engineering (3rd year student)
2023-Present (Current Student)
Expected to be graduated in 2027
GPA: 2.61/4

LANGUAGES

- Turkish (Working Proficiency)
- English (Fluent)
- Arabic (Native)

ENGINEERING PROJECTS

AUTONOMOUS SYSTEMS DEVELOPER

Academic Research Project: Autonomous Solar Inspection UAV

- Led the avionics integration and autopilot configuration for a quadcopter designed for autonomous solar farm inspection.
- Developed and deployed an offboard mission control architecture utilizing an Nvidia Jetson companion computer.
- Enabled real-time image processing and autonomous navigation by interfacing the Jetson with the flight controller to identify obstructed panels for targeted maintenance.

TEAM CAPTAIN & FLIGHT SOFTWARE DEVELOPER

Teknofest Fighter UAV 2025 | JETDOSE Team (9 members)

- Directed a 9-member cross-functional engineering team, overseeing the integration of avionics, flight software, and airframe mechanics.
- Engineered and validated autonomous Dogfight and Kamikaze mission logic using a custom Gazebo SITL (Software-In-The-Loop) environment
- Executed complex telemetry monitoring and mission commands as the Ground Control Station (GCS) operator, successfully advancing the platform to the competition semi-finals.

UAV SYSTEMS INTEGRATOR

Independent Project: 3D-Printed Tailsitter VTOL

- Manufactured and integrated a 3D-printed Tailsitter VTOL airframe to gain practical experience with dual-motor platform.
- Configured the avionics and flight control architecture using ArduPilot.
- Calibrated and tuned PID parameters and transition logic within ArduPilot to achieve stable flight during the hover, and transition.